

Distributions

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1 Binomial Distribution

Definition : A binomial distribution describes how many successes you get in a fixed number of trials when every trial has only two outcomes and the probability of success stays the same each time.

That is it. It counts how many times something happens out of a set number of attempts. Examples are coin flips, correct answers on a multiple choice quiz where each question has the same chance of being answered correctly, or how many customers out of 20 make a purchase. **Problem:**

A fair coin has a probability $p = \frac{1}{2}$ of landing heads. You flip the coin $n = 5$ times. What is the probability of getting exactly $k = 3$ heads?

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}$$

Solution:

First, identify the parameters:

$$n = 5, \quad k = 3, \quad p = \frac{1}{2}$$

Substitute them into the binomial formula:

$$P(X = 3) = \binom{5}{3} \left(\frac{1}{2}\right)^3 \left(1 - \frac{1}{2}\right)^{5-3}$$

Compute the combination term:

$$\binom{5}{3} = \frac{5!}{3!2!} = \frac{120}{6 \times 2} = 10$$

Now compute the probability pieces:

$$\begin{aligned} \left(\frac{1}{2}\right)^3 &= \frac{1}{8} \\ \left(1 - \frac{1}{2}\right)^2 &= \left(\frac{1}{2}\right)^2 = \frac{1}{4} \end{aligned}$$

Multiply all components:

$$P(X = 3) = 10 \times \frac{1}{8} \times \frac{1}{4}$$

$$P(X = 3) = 10 \times \frac{1}{32}$$

$$P(X = 3) = \frac{10}{32}$$

Simplify the fraction:

$$P(X = 3) = \frac{5}{16}$$

Final Answer:

$$P(X = 3) = \frac{5}{16}$$

2 Poisson Distribution

Definition.

A Poisson distribution describes the probability of observing a certain number of events in a fixed amount of time, space, or area when those events occur independently and at a constant average rate. It is used when events are rare, random, and do not influence each other.

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where:

X = number of events

k = the specific number of events you want the probability for

λ = the average number of events in the time interval

$$e \approx 2.71828$$

$$k! = k \times (k - 1) \times (k - 2) \times \cdots \times 1$$

Problem.

Cars pass a mailbox at an average rate of $\lambda = 3$ cars per minute. Find the probability that exactly $k = 5$ cars pass in one minute.

Poisson formula.

If X is Poisson with parameter λ , then

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}.$$

Given. $\lambda = 3, k = 5$.

$$P(X = 5) = \frac{3^5 e^{-3}}{5!}.$$

Compute each piece:

$$3^5 = 3 \times 3 \times 3 \times 3 \times 3 = 243,$$

$$5! = 5 \times 4 \times 3 \times 2 \times 1 = 120,$$

so

$$P(X = 5) = \frac{243 e^{-3}}{120}.$$

Using $e^{-3} \approx 0.049787068$,

$$P(X = 5) \approx \frac{243 \times 0.049787068}{120} = \frac{12.099}{120} \approx 0.10083.$$

Answer.

$$P(X = 5) \approx 0.1008 \quad (\text{about } 10.08\%).$$

3 Normal Distribution

Definition.

A normal distribution is a continuous probability distribution where values cluster around a central mean. Most observations lie close to the mean, and the probability decreases symmetrically as you move away from the mean. Its graph is a bell-shaped curve.

The probability density function is:

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Where:

$$\mu = \text{mean}, \quad \sigma = \text{standard deviation}$$

$$x = \text{value being evaluated}$$

Example.

Suppose a group of students has heights that follow a normal distribution with $\mu = 150$ cm and $\sigma = 10$ cm. Find the value of the density function at $x = 160$ cm.

$$\begin{aligned} f(160) &= \frac{1}{10\sqrt{2\pi}} \exp\left(-\frac{(160-150)^2}{2 \cdot 10^2}\right) \\ &= \frac{1}{10\sqrt{2\pi}} \exp\left(-\frac{100}{200}\right) \\ &= \frac{1}{10\sqrt{2\pi}} \exp(-0.5) \\ \exp(-0.5) &\approx 0.60653 \\ f(160) &\approx \frac{0.60653}{10\sqrt{2\pi}} \approx 0.0242 \end{aligned}$$

Final answer:

$$f(160) \approx 0.0242$$

4 Exponential Distribution

Definition.

An exponential distribution describes the time between independent events that happen at a constant average rate. It is used when you want to model waiting times such as time until a machine fails, time until a bus arrives, or time between phone calls.

Its probability density function is:

$$f(x) = \lambda e^{-\lambda x} \quad \text{for } x \geq 0$$

Where:

$$\lambda = \text{rate parameter}$$

$$x = \text{waiting time}$$

$$e \approx 2.71828$$

Example.

Suppose calls arrive at a help desk at an average rate of $\lambda = 0.2$ calls per minute. Find the probability density at $x = 5$ minutes.

$$f(x) = \lambda e^{-\lambda x}$$

Substitute the values:

$$f(5) = 0.2 e^{-0.2 \cdot 5}$$

$$= 0.2 e^{-1}$$

$$e^{-1} \approx 0.3679$$

So:

$$f(5) \approx 0.2 \times 0.3679 = 0.07358$$

Final answer:

$$f(5) \approx 0.0736$$

5 Difference

Distribution	What it models	Where it is used
Binomial	Number of successes in a fixed number of trials where each trial has two outcomes	A/B testing, classification accuracy, surveys, quality control
Poisson	Number of events occurring in a fixed time or space when events are rare and independent	Counting calls per hour, number of accidents per day, web traffic events
Normal	Continuous data centered around a mean with symmetric spread	Heights, test scores, measurement errors, manufacturing tolerances
Exponential	Time between independent events occurring at a constant rate	Waiting times, time until failure, queue systems, time between arrivals